

A Large-Scale Dataset of Distributed Renewable Energy Scenarios on the IEEE-33 bus network

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Subject: Research Methodology

Group 15

Problem Statement

How can we transform a raw, noisy, and high-dimensional distributed renewable energy dataset into a statistically consistent, reliable, and interpretable dataset using systematic research methodologies such as hypothesis testing, anomaly detection, and dimensionality reduction?

Literature Review

- Modern power systems increasingly rely on **distributed renewable energy (DRE)**, introducing high variability and uncertainty. [1] [2]
- Advanced forecasting and optimization methods using **deep learning and multi-energy systems** have been widely explored. [3] [4]
- Standard test systems like the **IEEE-33 bus network** provide a benchmark for distribution studies but lack realistic renewable datasets. [5]
- Existing datasets either focus on **centralized generation** or lack **spatiotemporal correlations and labeling**, limiting their usability. [6]

Research Gap

From the literature review, it is evident that there is a lack of datasets specifically designed for distributed renewable energy systems. Existing datasets do not adequately model spatial and temporal correlations and often suffer from missing values and lack of labeling. Additionally, integrated wind and solar scenarios are not sufficiently represented. This highlights a clear research gap, motivating the need for a statistically robust and well-structured dataset.

Base Paper

- Published in **2026**, solving **real world problem** for distributed renewable energy systems.
- Uses statistical preprocessing and synthetic data generation.
- Validates data using correlation and distribution-based methods.

DDRE-33 Dataset

- A large-scale dataset for distributed renewable energy scenarios on the **IEEE-33** bus system. [7]
- Captures spatial and temporal variations in wind and solar generation.
- Publicly available dataset released by the authors via repository. [8]

DataSet Overview

- DDRE-33 dataset for distributed renewable energy systems.
- Includes two energy sources:
 - Wind power
 - Solar (Photovoltaic) power
- Data collected from real-world distributed systems

Dataset Structure

- 2000 single-node time series
- 300 wind–PV combined scenarios
- Recorded at 15-minute intervals over one year

Dataset Attributes



timestamp	# PV_power_1	# PV_power_2	# PV_power_3	# PV_power_4	# PV_power_5	# PV_power_6	# PV_power_7	# PV_power_8
2022-07-01 00:00:00	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
2022-07-01 00:15:00	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
2022-07-01 00:30:00	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
2022-07-01 00:45:00	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
2022-07-01 01:00:00	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
2022-07-01 01:15:00	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
2022-07-01 01:30:00	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
2022-07-01 01:45:00	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
2022-07-01 02:00:00	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
2022-07-01 02:15:00	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
2022-07-01 02:30:00	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

Data of Node18PV

Data Representation

- Power represented in:
 - Megawatt (MW) – actual values
 - Per-unit (p.u.) – normalized values

$$p.u. = \frac{\text{Actual Power}}{\text{Maximum Capacity}} \quad (1)$$

Key Features

- Captures spatial and temporal variations in renewable generation
- Labeled scenarios based on weather and generation patterns
- Clean dataset: no missing values or anomalies after preprocessing
- Suitable for ML, simulation, and power system analysis

Outlier Detection

Methods Used

- Interquartile Range (IQR)

Interquartile Range (IQR) Formula

$$IQR = Q_3 - Q_1 \quad (2)$$

$$\text{Lower} = Q_1 - 1.5 \times IQR \quad (3)$$

$$\text{Upper} = Q_3 + 1.5 \times IQR \quad (4)$$

Impact

- **Noise Reduction:** Removes extreme anomalies.
- **Model Performance:** Improves overall prediction accuracy.

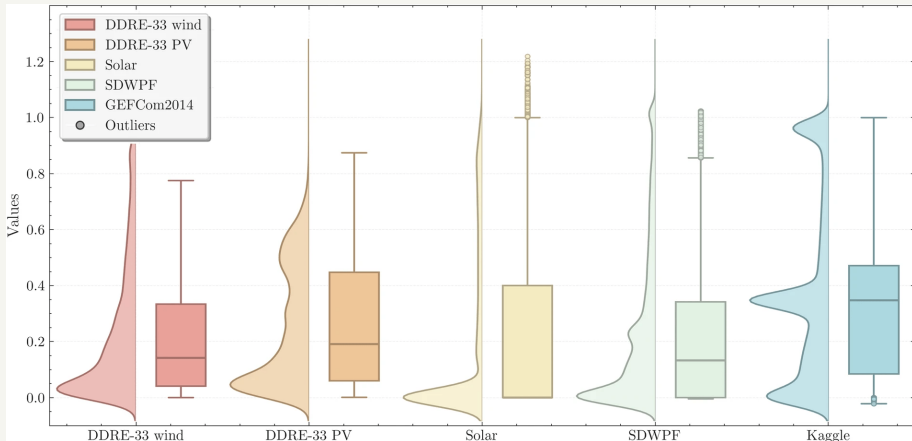


Figure: Comparison of power output distributions across datasets highlighting anomaly removal and improved data quality in DDRE-33

Missing Data Handling Methods Used

- Linear interpolation and Time-based averaging

Mathematical Formulation

$$x_t = x_{t_1} + \frac{(x_{t_2} - x_{t_1})}{(t_2 - t_1)} \cdot (t - t_1) \quad (5)$$

$$x_t = \frac{1}{n} \sum_{i=1}^n x_i \quad (6)$$

Impact

- **Continuity:** Maintains time-series flow.
- **Integrity:** Prevents distortion in temporal analysis.

Temporal Autocorrelation

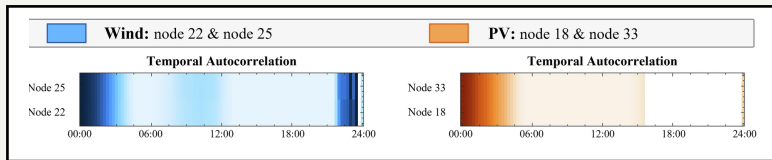
$$R_{xx}(\tau) = \frac{1}{N} \sum_{t=1}^{N-\tau} x_t x_{t+\tau}$$

- Measures similarity of a signal with itself over time lag τ

Spatial Cross-Correlation

$$R_{xy} = \frac{1}{N} \sum_{t=1}^N x_t y_t$$

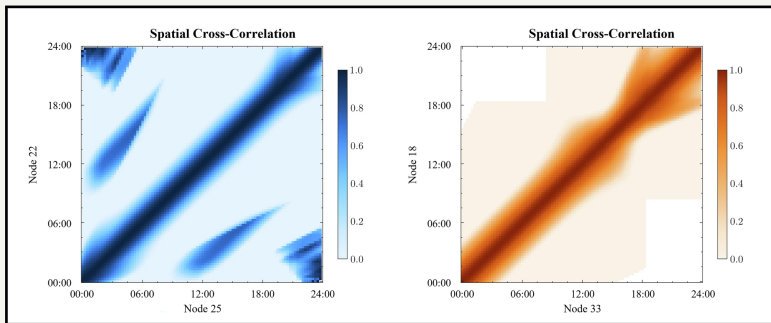
- Measures similarity between two nodes/signals



Temporal Correlation Analysis

Inference

- Dark colors at low lag indicate that consecutive time points are highly similar.
- Correlation decreases with increasing lag.
- Solar correlation drops to near zero at higher lag.



Spatial Correlation Analysis

Inference

- Dark diagonal patterns indicate high similarity between nodes at the same time.
- Consistent behavior in nearby locations.

Hypothesis Testing: PV Node 18 vs PV Node 33



Objective: Compare solar (PV) power output across two nodes.

Dataset Details:

- PV Node 18 vs PV Node 33
- Samples: 17,520,000 (each)

Statistical Methods:

- Two-sample t-test (mean comparison)
- F-test (variance comparison)

Hypotheses:

T-test:

- $H_0: \mu_{18} = \mu_{33}$
- $H_1: \mu_{18} \neq \mu_{33}$

F-test:

- $H_0: \sigma_{18}^2 = \sigma_{33}^2$
- $H_1: \sigma_{18}^2 \neq \sigma_{33}^2$

Significance Level: $\alpha = 0.05$

Descriptive Statistics:

- Mean PV18: 0.05478 MW Mean PV33: 0.08215 MW
- Var PV18: 0.00991 Var PV33: 0.02227

Statistical Tests:

- T-test: $T = -638.60$, P-value $\approx 0 \Rightarrow$ Reject H_0
- F-test: $F = 0.4447$, P-value $\approx 0 \Rightarrow$ Reject H_0

Results: PV Node 18 vs PV Node 33

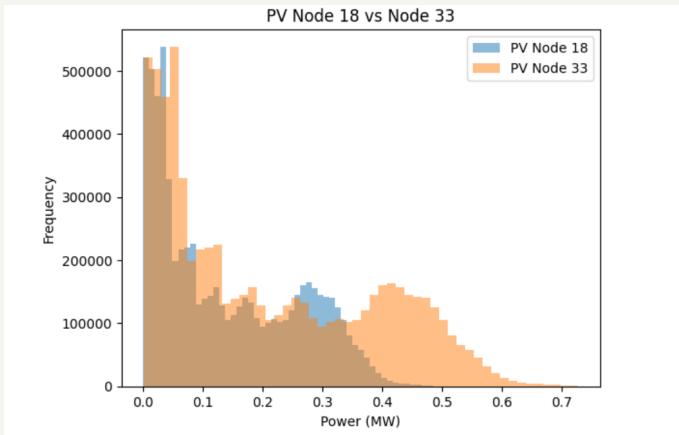


Fig : This histogram compares the power output distribution of PV Node 18 and 33.

Inference

- PV33 produces higher mean power than PV18
- Variability is also significantly higher at Node 33
- Results confirm statistical significance in both mean and variance

Conclusion

- Mean power outputs differ significantly (Reject H_0 in t-test)
- Variability differs significantly (Reject H_0 in F-test)
- Node 33 produces higher solar power than Node 18
- Node 33 exhibits greater variability (more fluctuation)
- Node 18 shows lower output but relatively more stable behavior

Wind Node 22 vs Node 25: Hypothesis & Method



Objective: Compare wind power output across two nodes.

Dataset:

- Node 22 vs Node 25 (Wind)
- Samples: 17,520,000 each

Statistical Tests:

- Two-sample t-test (mean comparison)
- F-test (variance comparison)

Hypotheses:

T-test:

- $H_0: \mu_{22} = \mu_{25}$
- $H_1: \mu_{22} \neq \mu_{25}$

F-test:

- $H_0: \sigma_{22}^2 = \sigma_{25}^2$
- $H_1: \sigma_{22}^2 \neq \sigma_{25}^2$

Significance Level: $\alpha = 0.05$

Descriptive Statistics:

- Mean W22: 0.03989 MW
- Mean W25: 0.02981 MW
- Var W22: 0.00182
- Var W25: 0.00096

T-Test:

- T-stat: 799.91
- P-value: 0.0

F-Test:

- F-stat: 1.9048
- P-value: ≈ 0

Decisions:

- Reject H_0 (T-test)
- Reject H_0 (F-test)

Wind Distribution: Node 22 vs Node 25

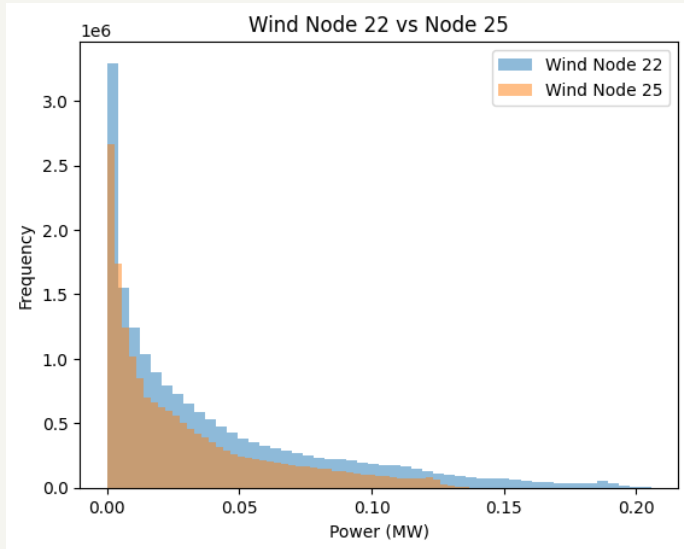


Fig : This histogram compares the power output distribution of Wind Node 22 and 25.

Inference

- Node 22 distribution extends further right → higher power output
- Node 25 concentrated at lower values → lower generation
- Both distributions are right-skewed
- Node 22 shows wider spread → higher variability

Conclusion

- Node 22 is more productive but slightly more fluctuating
- Node 25 is less productive but more stable

Why Anova?

- Multiple energy sources and locations:
 - Solar (Node 18, Node 33)
 - Wind (Node 22, Node 25)
- Need to compare **more than two groups simultaneously**

Why not T-test?

- T-test compares only **2 groups**
- Multiple T-tests increase **error**

Null Hypothesis (H_0):

- All energy sources have equal mean power output

Alternate Hypothesis (H_1):

- At least one group has a different mean

Methodology:

- Removed timestamp column
- Flattened scenario data (~ 17 M samples per group)
- Applied **One-Way ANOVA**

Results:

- F-statistic $\approx 1,038,981$
- p-value ≈ 0.0

Decision:

- $p < 0.05 \Rightarrow \text{Reject } H_0$

Interpretation:

- Significant differences exist across energy sources
- Solar $\rightarrow$ higher mean but more variable
- Wind $\rightarrow$ lower mean but more stable

Why?

- Labels are assigned based on rules
- Need to check:
 - Do they match real patterns?
 - Do different labels show different behavior?

Objective:

- Same label $\rightarrow$ similar data
- Different labels $\rightarrow$ different patterns

Impact:

- Improves data reliability
- Makes dataset useful for ML and analysis

Data Preparation:

- Time-series data converted into features
- High-dimensional features:
 - Wind $\rightarrow$ 35
 - Solar $\rightarrow$ 42

PCA:

$$Z = XW$$

- Projects data $\rightarrow$ 2D space
- Preserves maximum variance

Impact:

- Used for visualization & separability analysis

Method:

- K-means clustering applied
 - Wind → 4 clusters
 - Solar → 6 clusters

$$J = \sum_{i=1}^k \sum_{x \in C_i} \|x - \mu_i\|^2$$

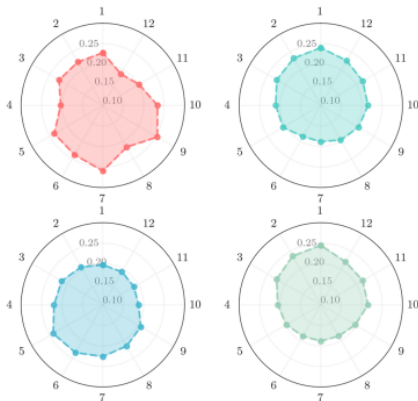
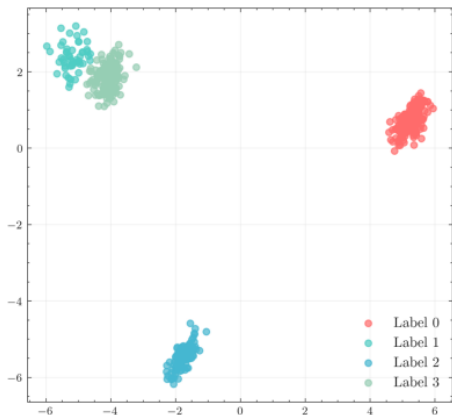
Objective:

- Group similar generation patterns

Impact:

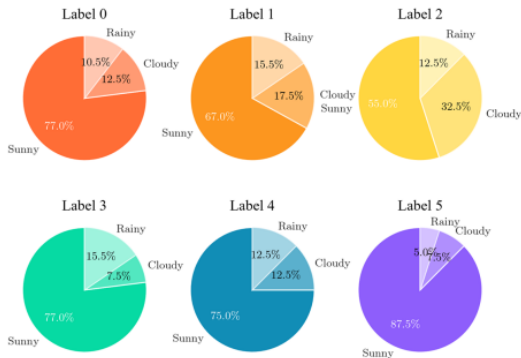
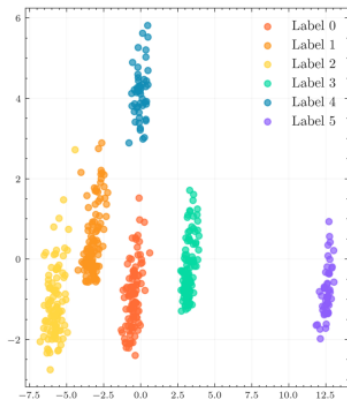
- Validates labels:
 - Same label → similar patterns
 - Different labels → distinct patterns

Clustering Results – Wind



- Each cluster shows distinct seasonal trends
- Label patterns: Label 0 → Spring/Autumn, Label 1 & 3 → Winter, Label 2 → Stable

Clustering Results – Solar



- Each cluster shows distinct weather-based patterns
- Label patterns: Label 0 & 5 → Mostly sunny, Label 1 & 2 → Mixed (cloudy/rainy), Label 3 & 4 → Moderate conditions

Applications

- Power planning and scheduling in distribution systems
- Power flow analysis under renewable conditions
- Machine learning for prediction and classification

Conclusion

- Clean and complete dataset
- Labels are meaningful and validated
- Suitable for analysis and research

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- [7] H. W. e. a. Chen Y., Xie H., "A large-scale dataset of distributed renewable energy scenarios on the ieee-33 bus network," *Scientific Data*, vol. 13, no. 146, 2026.

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- [9] Drishti, Muskaan, Priya, Pravallika, Saahitthi, and Shreya, "Solar and wind power variation analysis project," <https://github.com/pri2k/Solar-Wind-Power-Variation-Analysis-Project>, 2026, accessed: 2026-04-16.

Thank You